Product Optimization & Revenue Contribution Analysis for Afficionado Coffee Roasters

Abstract
This research investigates product-level sales and revenue dynamics within Afficionado Coffee Roasters using transaction-level retail data. The study evaluates product popularity, revenue contribution, category dependency, and revenue concentration using exploratory data analysis, KPI modeling, and Pareto analysis. Findings identify high-performing "hero" products, underperforming menu items, and operational inefficiencies affecting profitability. The project provides actionable recommendations for menu optimization, operational efficiency, and revenue growth.

1. Introduction

The specialty coffee retail industry depends heavily on product optimization, efficient menu engineering, and customer purchasing behavior analysis. Modern coffee retailers operate with diverse product portfolios including beverages, seasonal offerings, snacks, and premium specialty items. Afficionado Coffee Roasters maintains transaction-level sales data but lacked structured analytical insights into product-level performance, category dependence, and revenue optimization opportunities. This research aims to address those gaps using data-driven analytical techniques.

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2. Problem Statement

Despite maintaining detailed transactional records, the organization lacked: Visibility into product profitability Revenue concentration analysis Identification of underperforming products Category-level revenue dependency insights Data-driven menu optimization strategies Without structured analytics, operational decisions remained intuition-driven, increasing inefficiencies and limiting profitability optimization.

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3. Research Objectives

Primary Objectives Identify top-selling and least-selling products Measure revenue contribution by product and category Analyze revenue concentration across the menu **Secondary Objectives** Support menu simplification Identify high-impact "hero" products Improve operational efficiency Support strategic merchandising decisions

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4. Dataset Description

The dataset contains transaction-level information from Afficionado Coffee Roasters operations. Key variables include: transaction_id transaction_time transaction_qty unit_price store_location product_category product_type product_detail The dataset enables detailed product-level revenue and operational analysis.

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5. Methodology

The project followed a structured analytical workflow: **Data Cleaning** Removed duplicate records Validated pricing and quantities Standardized time formats **Revenue Computation** Revenue was calculated using:

Revenue = transaction_qty × unit_price

Exploratory Data Analysis Product popularity analysis Revenue contribution analysis Category performance analysis Store-level performance analysis Pareto analysis

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6. Product Popularity Analysis

Product popularity analysis measured total units sold by product and category. The study identified: Most frequently purchased products Least popular menu items Category demand distribution Store-level purchasing patterns High-volume products were compared against revenue generation to evaluate business impact.

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7. Revenue Contribution Analysis

Revenue contribution analysis evaluated the percentage of total revenue generated by each product and category. Key analytical outputs included: Revenue ranking by product Revenue share percentage Category-level revenue contribution Premium product profitability The analysis identified products contributing disproportionately to total revenue.

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8. Pareto Analysis

Pareto analysis was conducted to evaluate revenue concentration patterns across the menu. The study confirmed strong 80/20 behavior where a small percentage of products generated the majority of revenue. The analysis identified: Revenue anchor products Long-tail low-impact products Revenue concentration risks Menu imbalance patterns

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9. Key Findings

The project generated several important business insights: A limited number of products generated the majority of revenue Some high-volume products contributed relatively low profitability Several menu items produced minimal operational value Category-level revenue dependency created diversification risks Menu complexity increased operational inefficiency These findings support strategic menu engineering and product optimization initiatives.

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10. Strategic Recommendations

Menu Optimization Remove or redesign low-performing products Reduce menu clutter Prioritize high-margin products **Revenue Growth** Improve upselling strategies Bundle complementary products Promote premium beverages **Operational Efficiency** Simplify preparation workflows Improve inventory allocation Focus on standardized high-volume products

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11. Dashboard Development

A fully interactive Streamlit dashboard was developed to support business intelligence and operational decision-making. Dashboard capabilities include: KPI monitoring Revenue analysis Product popularity tracking Pareto visualization Store-level analytics Interactive filtering and drill-down analysis

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12. Business Impact

Implementation of the recommendations from this project can help Afficionado Coffee Roasters achieve: Improved revenue optimization Better menu engineering decisions Reduced operational inefficiency Improved inventory management Enhanced customer experience Long-term profitability improvement

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13. Conclusion

This research demonstrates how transaction-level product analytics can transform menu engineering and operational strategy within specialty coffee retail operations. Through structured data analysis, KPI modeling, Pareto evaluation, and dashboard development, the organization can transition from intuition-driven decisions toward a scalable data-driven operational framework. The findings provide a strong foundation for future predictive analytics, customer segmentation, and intelligent retail optimization.

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14. References

Retail Analytics and Business Intelligence Literature Menu Engineering Research Papers Pareto Principle Applications in Retail Coffee Industry Revenue Optimization Studies Operational Analytics and KPI Modeling Publications

Appendix A — Dataset Variables

Variable	Description
transaction_id	Unique transaction identifier
transaction_time	Time of transaction
transaction_qty	Quantity purchased

unit_price	Price per unit
store_location	Store branch location
product_category	Main product category
product_type	Product variation
product_detail	Detailed product name